

Scalability & Future Plan

Date	18 July 2026
Team ID	XXXXXX
Project Name	Smart Lender: Loan Eligibility Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local compute framework handles single evaluations sequentially, limiting concurrent pipeline requests.	High transaction traffic during heavy loan auditing windows experiences degradation and queue delays.	High
2	Flat structured file storage logs multi-row customer histories, which degrades over larger file parses.	Risk of write conflicts, slow historical analytical generation, and lack of relational index support.	Medium
3	Stateless local pipeline authentication model lacks decentralized role access boundaries control.	Restricts multi-branch institution deployments where credit analysts require strict segmented visibility.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles small manual local client test workloads	Migrate to multi-instance cloud clusters with auto-scaling to manage thousands of

		smoothly.	concurrent sessions.
2	Data Storage	Relies on local text directories and in-memory variables arrays.	Transition to fully enterprise-grade PostgreSQL or Cloud Datastore tables with relational indexes.
3	Performance	Local CPU handles parsing and evaluation workflows linearly.	Implement Redis memory caching strategies for frequently accessed scoring weights and parallel worker instances.
4	Security	Basic route checking without user tracking layers.	Integrate production OAuth2 authentication frameworks and role-based access control (RBAC) layers.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	OAuth2 integration and security dashboard views configuration.	Q3 2026	Ensures zero unauthorized model breaches during data transmissions.
Phase 2	Production relational database integration and active logs monitoring.	Q4 2026	Stabilizes dynamic retrieval timelines across multi-year portfolios.
Phase 3	Auto-scaling microservice clusters deployment and advanced model training.	Q1 2027	Supports cross-regional banking concurrent traffic seamlessly.